

# Steady State Error

## Chapter 7

# Steady state error from forward path transfer function

Steady state error is defined as

$$e(\infty) = \lim_{t \rightarrow \infty} e(t)$$

According to final value theorem

$$e(\infty) = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s)$$

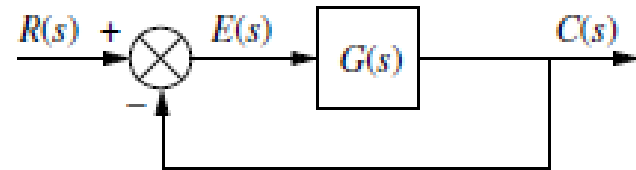
Deriving the expression for  $E(s)$  from block diagram of unity feedback system

$$E(s) = R(s) - C(s)$$

$$C(s) = E(s)G(s)$$

$$E(s) = \frac{R(s)}{1 + G(s)}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}$$



# Factors affecting steady state error by using the open loop transfer function

Steady state error

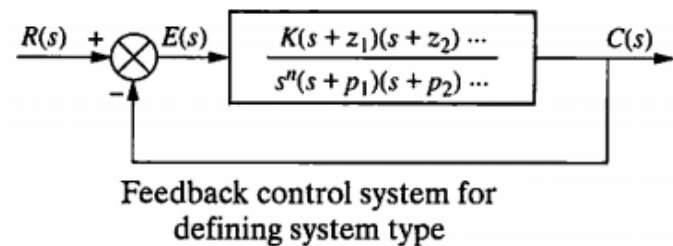
$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1 + G(s)}$$

1. input  $R(s)$
2. open loop transfer function  $G(s)$

# Type of the system

- Type of the system is defined to be the value of 'n' in the denominator or, equivalently, the number of pure integrators in the forward path

$$G(s) = \frac{(s + z_1)(s + z_2) \cdots}{s^n(s + p_1)(s + p_2) \cdots}$$



- If  $n = 0$ ,  $G(s)$  system is Type 0
- If  $n = 1$ ,  $G(s)$  system is Type 1
- If  $n = 2$ ,  $G(s)$  system is Type 2 and so on

# System Type

Identify the type of system for given forward path transfer function

a) 
$$G(s) = \frac{K(s+8)}{(s+7)(s+9)}$$

**Type 0**

b) 
$$G(s) = \frac{10(s+20)(s+30)}{s(s+25)(s+35)}$$

**Type 1**

c) 
$$G(s) = \frac{10(s+20)(s+30)}{s^2(s+25)(s+35)(s+50)}$$

**Type 2**

d) 
$$G(s) = \frac{K(2s+1)}{s^3}$$

**Type 3**

# Static Error Constants

- $K_p$ ,  $K_v$  and  $K_a$  are performance parameters of  $e_{ss}$  and these are called **static error constants**.

- Position error constant

$$K_p = \lim_{s \rightarrow 0} G(s)$$

- Velocity error constant

$$K_v = \lim_{s \rightarrow 0} sG(s)$$

- Acceleration error constant

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)$$

# Steady state error due to step input

- For steady state error to be zero

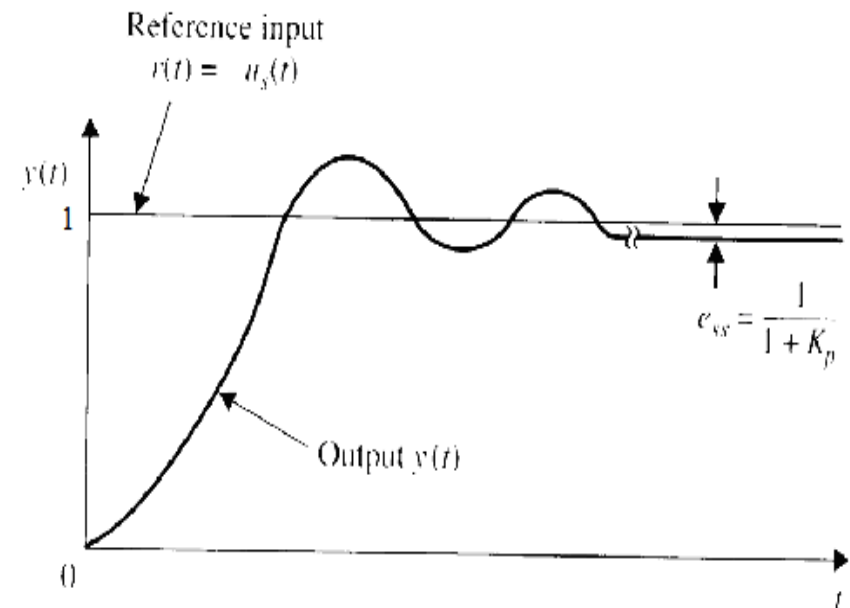
$$e(\infty) = \frac{1}{1 + K_p} = \frac{1}{\infty} = 0$$

$$K_p = \lim_{s \rightarrow 0} G(s) = \infty$$

- The values of static error constant depend upon the form of  $G(s)$  specially the number of pure integrators in the forward path.
- For the limit to be infinite, the denominator of  $G(s)$  must equal to zero as  $s$  goes to zero
- There must be atleast one integration in the forward path

$$G(s) = \frac{(s + z_1)(s + z_2) \cdots}{s (s + p_1)(s + p_2) \cdots}$$

- Type 1 and above for  $e_{ss}=0$



# Steady state error due to ramp input

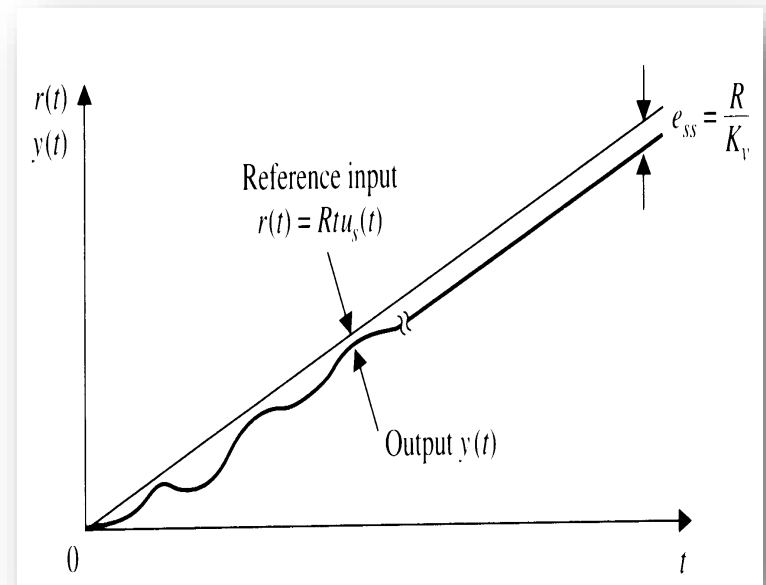
- For steady state error to be zero for ramp input

$$e(\infty) = \frac{1}{K_v} = \frac{1}{\infty} = 0 \quad K_v = \lim_{s \rightarrow 0} sG(s) = \infty$$

- For the limit to be infinite, the denominator of  $G(s)$  must equal to zero as  $s$  goes to zero
- There must be at least two integrations in the forward path

$$G(s) = \frac{(s + z_1)(s + z_2) \cdots}{s^2(s + p_1)(s + p_2) \cdots}$$

- Type 2 and above for  $e_{ss}=0$





# Steady state error due to parabolic input

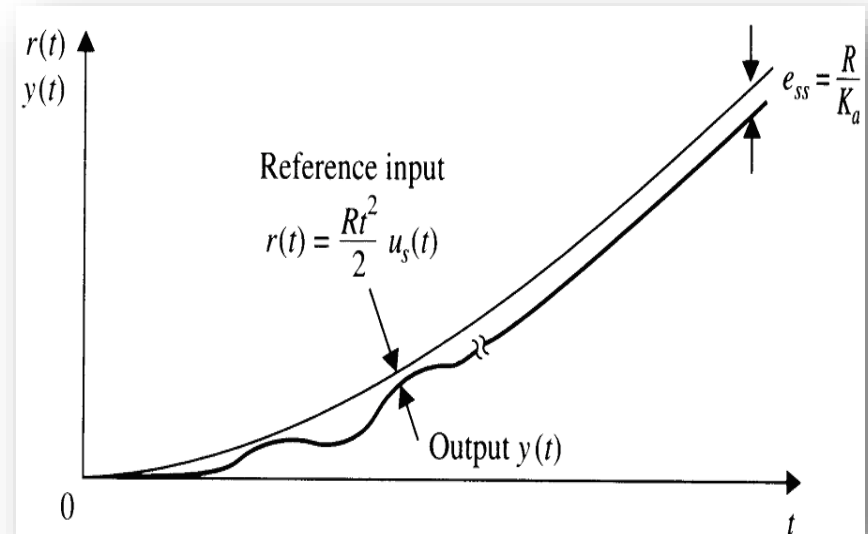
- For steady state error to be zero

$$e(\infty) = \frac{1}{K_a} = \frac{1}{\infty} = 0 \quad K_a = \lim_{s \rightarrow 0} s^2 G(s) = \infty$$

- For the limit to be infinite, the denominator of  $G(s)$  must equal to zero as  $s$  goes to zero
- There must be at least three integrations in the forward path

$$G(s) = \frac{(s + z_1)(s + z_2) \cdots}{s^3(s + p_1)(s + p_2) \cdots}$$

- Type 3 and above for  $e_{ss}=0$



**TABLE 7.2** Relationships between input, system type, static error constants, and steady-state errors

| Input                          | Steady-state<br>error formula | Type 0                   |                     | Type 1                   |                 | Type 2                   |                 |
|--------------------------------|-------------------------------|--------------------------|---------------------|--------------------------|-----------------|--------------------------|-----------------|
|                                |                               | Static error<br>constant | Error               | Static error<br>constant | Error           | Static error<br>constant | Error           |
| Step, $u(t)$                   | $\frac{1}{1 + K_p}$           | $K_p = \text{Constant}$  | $\frac{1}{1 + K_p}$ | $K_p = \infty$           | 0               | $K_p = \infty$           | 0               |
| Ramp, $tu(t)$                  | $\frac{1}{K_v}$               | $K_v = 0$                | $\infty$            | $K_v = \text{Constant}$  | $\frac{1}{K_v}$ | $K_v = \infty$           | 0               |
| Parabola, $\frac{1}{2}t^2u(t)$ | $\frac{1}{K_a}$               | $K_a = 0$                | $\infty$            | $K_a = 0$                | $\infty$        | $K_a = \text{Constant}$  | $\frac{1}{K_a}$ |

## Example 7.3

**PROBLEM:** Find the steady-state errors for inputs of  $5u(t)$ ,  $5tu(t)$ , and  $5t^2u(t)$  to the system shown in Figure 7.6. The function  $u(t)$  is the unit step.

Example 7.3

→ closed loop system is stable (Verify)

$$G(s) = \frac{100(s+2)(s+6)}{s(s+3)(s+4)}, \text{ Type 1 system}$$

(a)  $e_{ss} = ?$  when  $r(t) = 5u(t)$

$$R(s) = \frac{5}{s}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)} = \lim_{s \rightarrow 0} \frac{s \cdot 5/s}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{5}{1 + G(s)} = \frac{5}{1 + \lim_{s \rightarrow 0} G(s)}$$

$$\lim_{s \rightarrow 0} G(s) = \lim_{s \rightarrow 0} \frac{100(s+2)(s+6)}{s(s+3)(s+4)} = \infty$$

So,

$$e(\infty) = \frac{5}{1 + \infty} = \frac{5}{\infty} = 0$$

→ zero error for step input

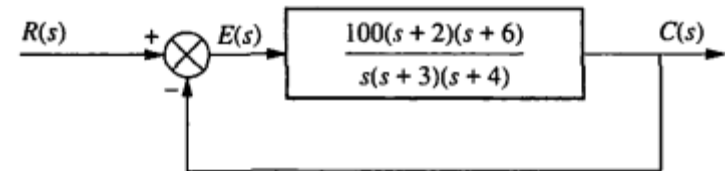
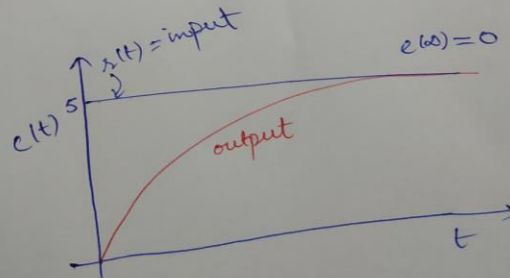


FIGURE 7.6 Feedback control system

b)  $e_{ss} = ?$  when  $r(t) = st u(t)$

$$R(s) = \frac{5}{s^2}, \text{ Ramp input}$$

$$e(\infty) = e_{\text{ramp}}(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)} = \lim_{s \rightarrow 0} \frac{s \times 5/s^2}{1+G(s)}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{5}{s+sG(s)} = \frac{5}{\lim_{s \rightarrow 0} [s+sG(s)]}$$

$$e(\infty) = \frac{5}{\lim_{s \rightarrow 0} sG(s)}$$

$$\lim_{s \rightarrow 0} sG(s) = \lim_{s \rightarrow 0} s \times \frac{100(s+2)(s+6)}{s(s+3)(s+4)} = \frac{100 \times 2 \times 6}{3 \times 4} = 100$$

$$e(\infty) = \frac{5}{100} = \frac{1}{20}$$

→ finite error for a ramp input



c)  $e(\infty) = e_{ss} = ?$  when  $r(t) = st^2 u(t)$ ,  $R(s) = \frac{10}{s^3}$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)} = \lim_{s \rightarrow 0} \frac{s \times 10/s^3}{1+G(s)}$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{10}{s^2+s^2G(s)} = \frac{10}{\lim_{s \rightarrow 0} s^2G(s)}$$

$$\lim_{s \rightarrow 0} s^2G(s) = \lim_{s \rightarrow 0} s^2 \times \frac{100(s+2)(s+6)}{s(s+3)(s+4)} = 0$$

$$e(\infty) = \frac{10}{0} = \infty$$

# Steady State error Specifications

- Static error constants can be used to specify the steady-state characteristics of control systems
  - Stable system
  - System type
  - Test signal
  - $e_{ss}$

## Interpreting the Steady-State Error Specification

**PROBLEM:** What information is contained in the specification  $K_p = 1000$ ?

**SOLUTION:** The system is stable. The system is Type 0, since only a Type 0 system has a finite  $K_p$ . Type 1 and Type 2 systems have  $K_p = \infty$ . The input test signal is a step, since  $K_p$  is specified. Finally, the error per unit step is

$$e(\infty) = \frac{1}{1 + K_p} = \frac{1}{1 + 1000} = \frac{1}{1001} \quad (7.5)$$

# Gain design to meet a steady state error specification

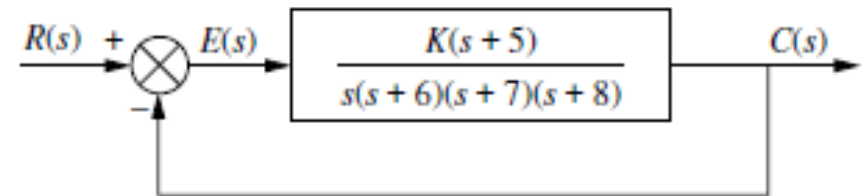
**PROBLEM:** Given the control system in Figure 7.10, find the value of  $K$  so that there is 10% error in the steady state.

$$e(\infty) = \frac{1}{K_v} = 0.1$$

$$K_v = 10 = \lim_{s \rightarrow 0} sG(s) = \lim_{s \rightarrow 0} s \times \frac{K(s+5)}{s(s+6)(s+7)(s+8)} = \frac{K \times 5}{6 \times 7 \times 8}$$

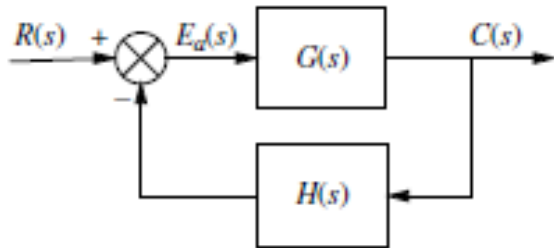
$$K = 672$$

Check if the system is stable at this value of  $K$ .

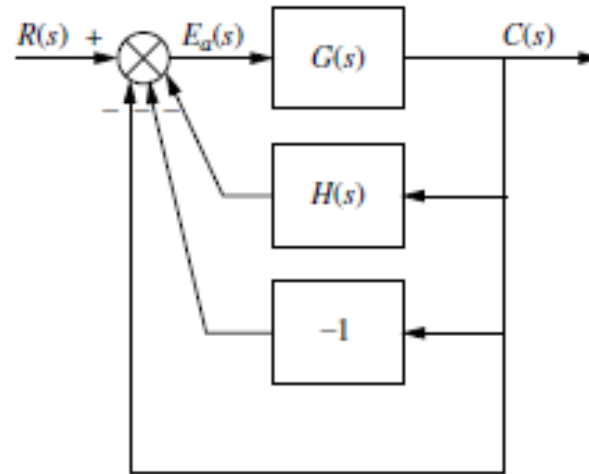


**FIGURE 7.10** Feedback control system for Example 7.6

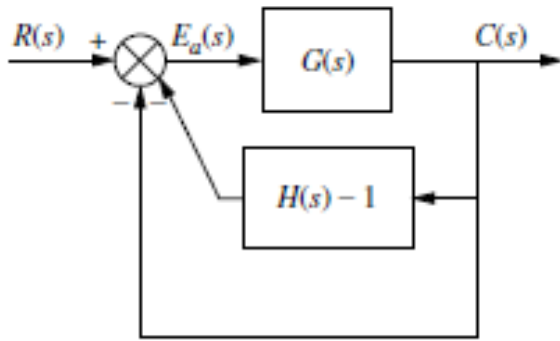
# Steady State error for non-unity feedback system



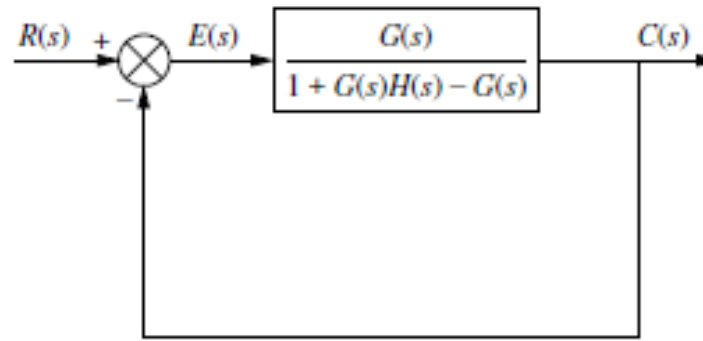
(b)



(c)



(d)



(e)

Actuating signal  $E_a(s)$ , is NOT the SSE  $E(s)$ .



**PROBLEM:** For the system shown in Figure 7.16, find the system type, the appropriate error constant associated with the system type, and the steady-state error for a unit step input. Assume input and output units are the same.

$$G(s) = \frac{100}{s(s+10)}$$

$$H(s) = \frac{1}{(s+5)}$$

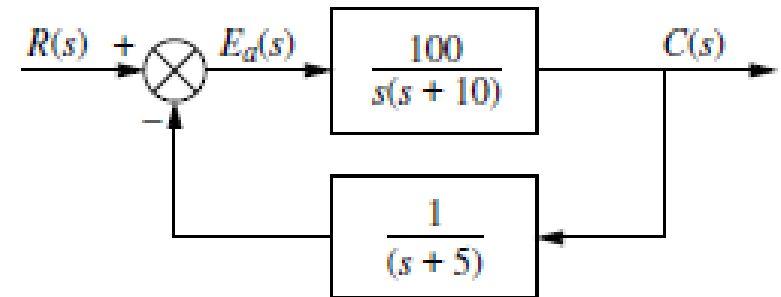
$$G_e(s) = \frac{G(s)}{1 + G(s)H(s) - G(s)}$$

$$G_e(s) = \frac{G(s)}{1 + G(s)H(s) - G(s)} = \frac{100(s+5)}{s^3 + 15s^2 - 50s - 400}$$

$$K_p = \lim_{s \rightarrow 0} G_e(s) = \frac{100 \times 5}{-400} = -\frac{5}{4}$$

$$e(\infty) = \frac{1}{1 + K_p} = \frac{1}{1 - (5/4)} = -4$$

Negative error means output is larger than the input.



**FIGURE 7.16** Nonunity feedback control system for Example 7.8

# Steady State error for Disturbances

$$C(s) = G_2(s) [G_1(s) E(s) + D(s)]$$

$$= G_1(s) G_2(s) E(s) + G_2(s) D(s)$$

$$E(s) = R(s) - C(s)$$

$$C(s) = R(s) - E(s)$$

$$R(s) - E(s) = G_1(s) G_2(s) E(s) + G_2(s) D(s)$$

$$[1 + G_1(s) G_2(s)] E(s) = R(s) - G_2(s) D(s)$$

$$E(s) = \frac{1}{1 + G_1(s) G_2(s)} R(s) - \frac{G_2(s)}{1 + G_1(s) G_2(s)} D(s)$$

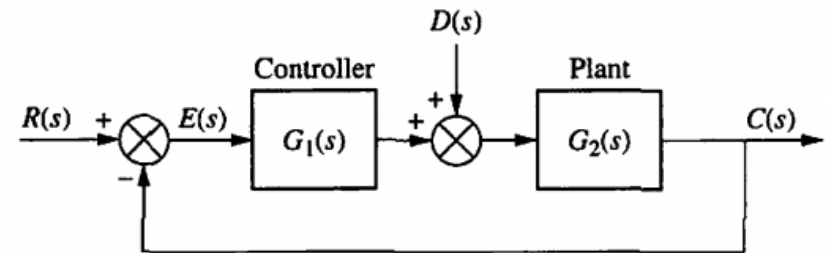
$$e(\infty) = \lim_{s \rightarrow 0} s E(s)$$

$$= \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G_1(s) G_2(s)} - \lim_{s \rightarrow 0} \frac{s G_2(s)}{1 + G_1(s) G_2(s)} D(s)$$

$$= e_R(\infty) + e_D(\infty)$$

where  $e_R(\infty) = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G_1(s) G_2(s)}$

$$e_D(\infty) = - \lim_{s \rightarrow 0} \frac{s G_2(s)}{1 + G_1(s) G_2(s)} D(s)$$



Feedback control system showing disturbance

# Steady State error for Disturbances

$$R(s)=0$$

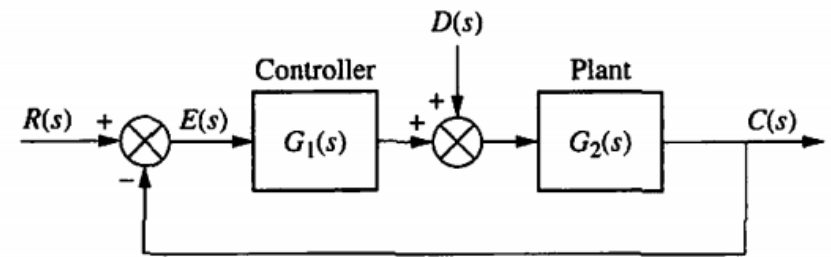
$$e_D(\infty) = -\lim_{s \rightarrow 0} \frac{sG_2(s)}{1 + G_1(s)G_2(s)} D(s)$$

$$\bar{D}(s) = 1/s$$

$$e_D(\infty) = -\frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)}$$

$$e_D(\infty) = -\frac{1}{\frac{1}{\lim_{s \rightarrow 0} G_2(s)} + \lim_{s \rightarrow 0} G_1(s)}$$

$$e_D(\infty) \downarrow \text{ as } \lim_{s \rightarrow 0} G_2(s) \downarrow \quad \lim_{s \rightarrow 0} G_1(s) \uparrow$$



Feedback control system showing disturbance

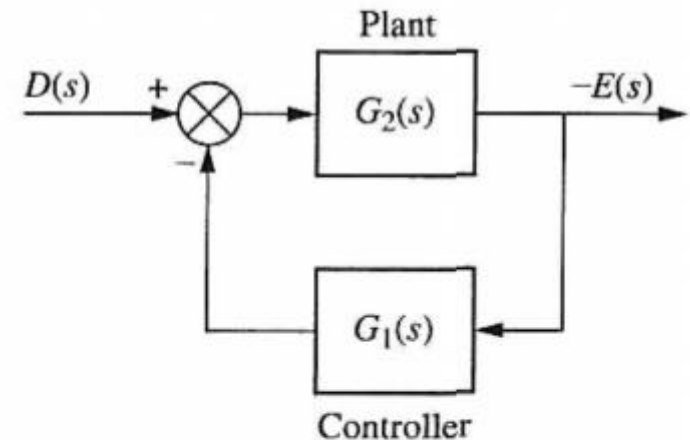


FIGURE 7.12 Figure 7.11 system rearranged to show disturbance as input and error as output, with  $R(s) = 0$

## Steady-State Error Due to Step Disturbance

**PROBLEM:** Find the steady-state error component due to a step disturbance for the system of Figure 7.13.

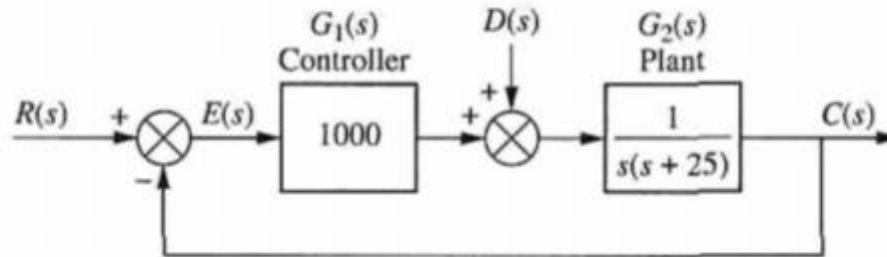
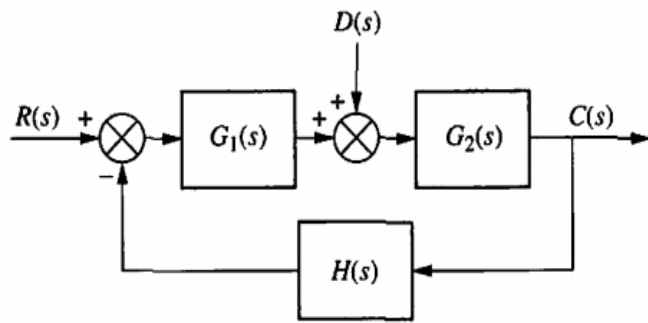


FIGURE 7.13 Feedback control system for Example 7.7

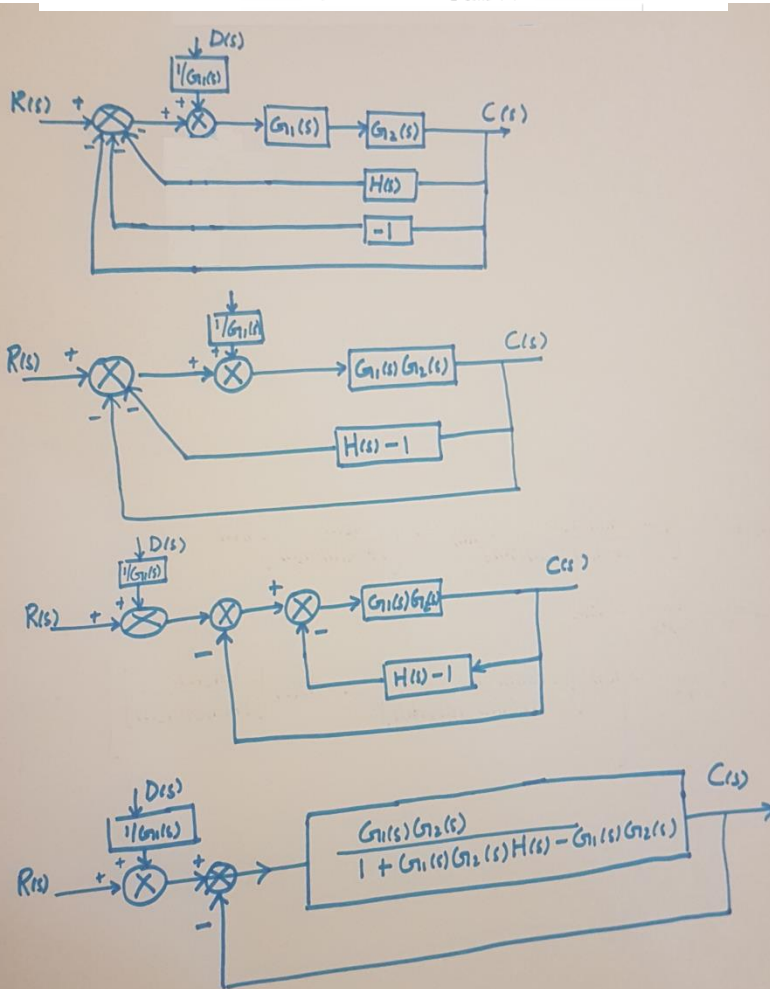
$$e_D(\infty) = -\frac{1}{\lim_{s \rightarrow 0} \frac{1}{G_2(s)} + \lim_{s \rightarrow 0} G_1(s)} = -\frac{1}{0 + 1000} = -\frac{1}{1000}$$

The result shows that the steady-state error produced by the step disturbance is inversely proportional to the dc gain of  $G_1(s)$ . The dc gain of  $G_2(s)$  is infinite in this example.



Nonunity feedback control system with disturbance

# Non-Unity Feedback Control System with disturbance



$$C(s) = \left[ \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)} \right] \left[ E(s) + \frac{D(s)}{G_1(s)} \right]$$

$$E(s) = R(s) - C(s)$$

$$= R(s) - \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)} E(s) - \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)} \frac{D(s)}{G_1(s)}$$

$$E(s) \left[ 1 + \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)} \right] = R(s) - \frac{G_2(s)}{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)} D(s)$$

$$E(s) \left[ \frac{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s) + G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)} \right] = R(s) - \frac{G_2(s) D(s)}{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)}$$

$$E(s) = \frac{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s)} R(s) - \frac{1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s)} \frac{G_2(s) D(s)}{[1 + G_1(s) G_2(s) H(s) - G_1(s) G_2(s)]}$$

$$E(s) = \left[ 1 - \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s)} \right] R(s) - \left[ \frac{G_2(s)}{1 + G_1(s) G_2(s)} \right] D(s)$$

$$e(\infty) = \lim_{s \rightarrow 0} s E(s)$$

$$= \lim_{s \rightarrow 0} s \left\{ R(s) \left[ 1 - \frac{G_1(s) G_2(s)}{1 + G_1(s) G_2(s) H(s)} \right] - D(s) \left[ \frac{G_2(s)}{1 + G_1(s) G_2(s)} \right] \right\}$$

# Steady state error of non unity feedback control system with disturbance

$$e(\infty) = \lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} s \left\{ \left[ 1 - \frac{G_1(s)G_2(s)}{1 + G_1(s)G_2(s)H(s)} \right] R(s) - \left[ \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} D(s) \right] \right\}$$

$$R(s) = D(s) = 1/s$$

$$e(\infty) = \lim_{s \rightarrow 0} sE(s) = \left\{ \left[ 1 - \frac{\lim_{s \rightarrow 0} [G_1(s)G_2(s)]}{\lim_{s \rightarrow 0} [1 + G_1(s)G_2(s)H(s)]} \right] - \left[ \frac{\lim_{s \rightarrow 0} G_2(s)}{\lim_{s \rightarrow 0} [1 + G_1(s)G_2(s)H(s)]} \right] \right\}$$

For zero error,

$$\frac{\lim_{s \rightarrow 0} [G_1(s)G_2(s)]}{\lim_{s \rightarrow 0} [1 + G_1(s)G_2(s)H(s)]} = 1 \quad \text{and} \quad \frac{\lim_{s \rightarrow 0} G_2(s)}{\lim_{s \rightarrow 0} [1 + G_1(s)G_2(s)H(s)]} = 0$$

The two equations can always be satisfied if

- (1) the system is stable,
- (2)  $G_1(s)$  is a Type 1 system,
- (3)  $G_2(s)$  is a Type 0 system, and
- (4)  $H(s)$  is a Type 0 system with a dc gain of unity

# Steady state error from Closed loop transfer function

- From def. of Steady State error

$$e(\infty) = \lim_{t \rightarrow \infty} e(t)$$

- According to final value theorem

$$e(\infty) = \lim_{s \rightarrow 0} sE(s)$$

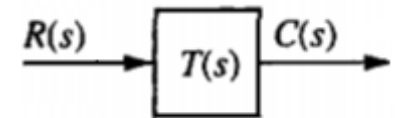
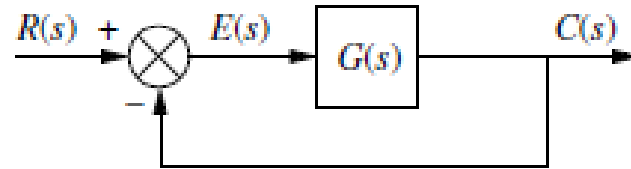
- Deriving the expression for E(s) of unity feedback system

$$E(s) = R(s) - C(s)$$

$$C(s) = R(s)T(s)$$

$$E(s) = R(s)[1 - T(s)]$$

$$e(\infty) = \lim_{s \rightarrow 0} sR(s)[1 - T(s)]$$



In Book

$$e(\infty) = \lim_{s \rightarrow \infty} sR(s)[1 - T(s)] \quad \text{X}$$

- This allows the steady-state error to be determined for a given **test input, R(s)** and the **closed loop transfer function, T(s)** of the system.

**PROBLEM:** Find the steady-state error for the system if  $T(s) = 5/(s^2 + 7s + 10)$  and the input is a unit step.

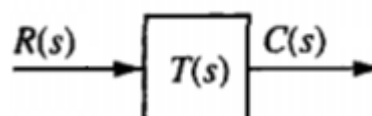
**SOLUTION:** From the problem statement,  $R(s) = 1/s$  and  $T(s) = 5/(s^2 + 7s + 10)$ .

$$E(s) = R(s)[1 - T(s)]$$

$$E(s) = \frac{s^2 + 7s + 5}{s(s^2 + 7s + 10)}$$

$$e(\infty) = \lim_{s \rightarrow 0} sE(s)$$

$$e(\infty) = 1/2.$$





# Stability in State Space

$$E(s) = R(s)[1 - T(s)] \quad (7.87)$$

Using Eq. (3.73) for  $T(s)$ , we find

$$E(s) = R(s)[1 - \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}] \quad (7.88)$$

Applying the final value theorem, we have

$$\lim_{s \rightarrow 0} sE(s) = \lim_{s \rightarrow 0} sR(s)[1 - \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}] \quad (7.89)$$

Let us apply the result to an example.

## Example 7.13

### Steady-State Error Using the Final Value Theorem

**PROBLEM:** Evaluate the steady-state error for the system described by Eqs. (7.90) for unit step and unit ramp inputs. Use the final value theorem.

$$\mathbf{A} = \begin{bmatrix} -5 & 1 & 0 \\ 0 & -2 & 1 \\ 20 & -10 & 1 \end{bmatrix}; \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}; \quad \mathbf{C} = [-1 \quad 1 \quad 0] \quad (7.90)$$

**SOLUTION:** Substituting Eqs. (7.90) into (7.89), we obtain

$$\begin{aligned} e(\infty) &= \lim_{s \rightarrow 0} sR(s) \left( 1 - \frac{s+4}{s^3 + 6s^2 + 13s + 20} \right) \\ &= \lim_{s \rightarrow 0} sR(s) \left( \frac{s^3 + 6s^2 + 12s + 16}{s^3 + 6s^2 + 13s + 20} \right) \end{aligned} \quad (7.91)$$

For a unit step,  $R(s) = 1/s$ , and  $e(\infty) = 4/5$ . For a unit ramp,  $R(s) = 1/s^2$ , and  $e(\infty) = \infty$ . Notice that the system behaves like a Type 0 system.

### TryIt 7.5

Use MATLAB, the Symbolic Math Toolbox, and the following statements to find the steady-state error for a step input to the system of Example 7.13.

```
syms s
A=[-5 1 0
    0 -2 1
    20 -10 1];
B=[0;0;1];
C=[-1 1 0];
I=[1 0 0
    0 1 0
    0 0 1];
E=(1/s)*[1-C*...
    [(s*I-A)^-1]*B];
%New command:
%subs(X,old,new):
%Replace old in ...
%X(old) with new.
error=subs(s*I-E,s,0)
```